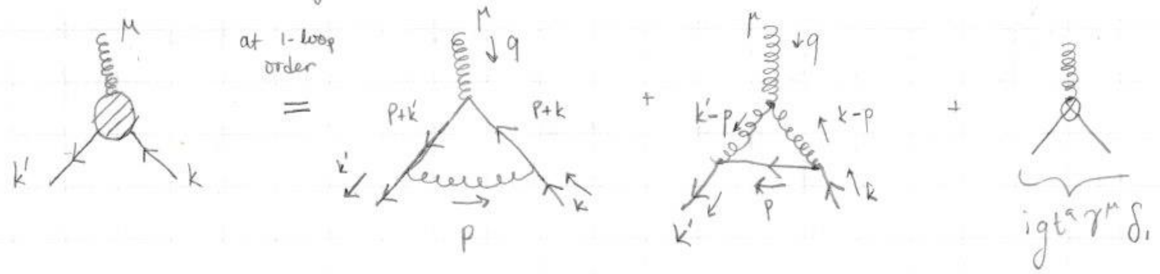
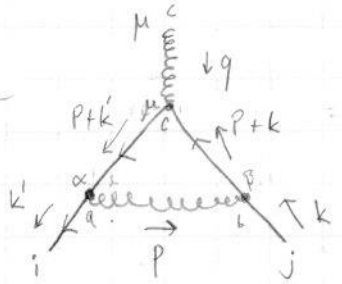
Quntum Field theory IV Week 3 Homework

Exercise 3.8

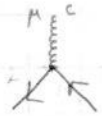
Compute the divergent part of 1-loop contribution to the fermion gluon vertex.

$-i\Gamma^\mu =$ 

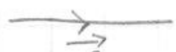
Let's start by calculating the first diagram of the three:



here μ, α, β are Lorentz indices, a, b, c, are gluon indices, and l, i, j, r are colour indices. Now let's use Feynman-rules from Peskin chapter A.1. We have the following Feynman rules; (We are using Feynman-'t Hooft gauge)

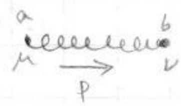
vertex:  $= ig\gamma^\mu t^c$,

Since we are only interested in the loop integral there is need for external spinor or polarization vectors.

the fermion propagator  is

$$\frac{i(\not{p}+m)}{p^2-m^2} \times \mathbb{1}_{colour}$$

where $\mathbb{1}_{colour}$ is indentity matrix in colour space

and similarily the gluon propagator  is

$$\frac{-i g_{\mu\nu}}{p^2} \delta_{ab}$$

Using these Feynman-rules we get the following loop integral in D-dimension:

$$I_1 = \mu^\varepsilon \int \frac{d^Dp}{(2\pi)^D} \left(ig\gamma^\alpha t^a\right) \frac{i(\not{p}+\not{k}'+m)}{(p+k')^2-m^2} \left(ig\gamma^\mu t^c\right) \frac{i(\not{p}+\not{k}+m)}{(p+k)^2-m^2} \left(ig\gamma^\beta t^b\right) \frac{-ig_{\alpha\beta}}{p^2}\delta_{ab}$$

Since we are interested in the UV-divergence we can only look at $p \gg \{k, m, k'\}$ range where we can do approximations where $\{k, k', m\} \to 0$. so we get $k = k' = m = 0$

This gives us

$$I_1 = \mu^\varepsilon g^3 \left(t^a t^c t^a\right) \int \frac{d^Dp}{(2\pi)^D} \frac{\gamma^\alpha \not{p}\gamma^\mu \not{p}\gamma_\alpha}{p^2\, p^2\, p^2} = \mu^\varepsilon g^3 \left(t^a t^c t^a\right) \int \frac{d^Dp}{(2\pi)^D} \frac{p^\rho p^\sigma \gamma^\alpha \gamma_\rho \gamma^\mu \gamma_\sigma \gamma_\alpha}{p^6}$$

Symmetry allows us to replace $p^\mu p^\nu \to \frac{1}{D} p^2 g^{\mu\nu}$ in the dimensional integral (Peskin eq.(A.41)). Then we have the following integral:

$$I_1 = \mu^\varepsilon t^a t^c t^a g^3 \int \frac{d^D p}{(2\pi)^D} \frac{1}{D} \frac{\gamma^\alpha \gamma^\rho \gamma^\mu \gamma_\rho \gamma_\alpha}{p^4}$$

Now we use the properties of Dirac algebra in D-dimension [Peskin eq. (A.55)]:

$$\gamma^\rho \gamma^\mu \gamma_\rho = -(d-2)\gamma^\mu \quad \Rightarrow \quad \gamma^\alpha \gamma^\rho \gamma^\mu \gamma_\rho = (d-2)^2 \gamma^\mu$$

$D = d =$ dimension (unknowingly changed the notation when reading Peskin equations)

$$\Rightarrow I_1 = \mu^\varepsilon (t^a t^c t^a) g^3 \frac{(d-2)^2}{d} \gamma^\mu \int \frac{d^D p}{(2\pi)^D} \frac{1}{(p^2)^2}$$

, now let's add a a small regulator Λ to compute this loop integral

$$I_1 = \mu^\varepsilon (t^a t^c t^a) g^3 \frac{(d-2)^2}{d} \gamma^\mu \int \frac{d^D p}{(2\pi)^D} \frac{1}{(p^2-\Lambda)^2}$$

Now let's use the Minkowski integral of Peskin eq. (A.44) $d = 4-\varepsilon$

$$I_1 = \mu^\varepsilon (t^a t^c t^a) g^3 \frac{(d-2)^2}{d} \gamma^\mu \frac{i}{(4\pi)^{d/2}} \frac{\Gamma(2-d/2)}{\Gamma(2)} \left(\frac{1}{\Lambda}\right)^{2-d/2}$$

$$I_1 = i g^3 \mu^\varepsilon \gamma^\mu t^a t^c t^a \frac{\Gamma(\varepsilon/2)}{(4\pi)^2} (4\pi)^{\varepsilon/2} \frac{(2-\varepsilon)^2}{4-\varepsilon} \left(\frac{1}{\Lambda}\right)^{\varepsilon/2}$$

$$I_1 = \frac{i g^3}{(4\pi)^2} \gamma^\mu t^a t^c t^a \left(\frac{2}{\varepsilon} - \gamma_E + O(\varepsilon)\right) \left[1 + \frac{\varepsilon}{2}\ln(4\pi) - \frac{\varepsilon}{2}\ln\left(\frac{\Lambda}{\mu^2}\right)\right] \frac{1}{4} \frac{(2-\varepsilon)^2}{(1+\frac{\varepsilon}{4})^{-1}}$$

$$\Gamma(\varepsilon/2) = \frac{2}{\varepsilon} - \gamma_E + O(\varepsilon)$$

$$x^{\varepsilon/2} = \exp(\varepsilon/2 \ln(x)) \simeq 1 + \frac{\varepsilon}{2}\ln(x)$$

$$\frac{1}{4-\varepsilon} = \frac{1}{4}(1-\varepsilon/4)^{-1} \simeq \frac{1}{4}\left(1+\frac{\varepsilon}{4}\right)$$

↑ Binomial approximation

Where we have dropped many terms of order ε^2 but not all of them. We can further write:

$$(2-\varepsilon)^2 (1+\varepsilon/4) = (4 - 2\varepsilon + \varepsilon^2)(1+\varepsilon/4) \simeq 4 - 2\varepsilon + \varepsilon = 4 - \varepsilon$$

to order $O(\varepsilon)$ $\Rightarrow \frac{1}{4}(4-\varepsilon) = 1 - \varepsilon/4$

$$\Rightarrow I_1 = i \frac{g^3}{(4\pi)^2} \gamma^\mu t^a t^c t^a \left[\frac{2}{\varepsilon} - \gamma_E\right] \left(1 + \frac{\varepsilon}{2}\ln(4\pi) - \frac{\varepsilon}{2}\ln\left(\frac{\Lambda}{\mu^2}\right) - \frac{\varepsilon}{4} + O(\varepsilon^2)\right)$$

Now let's simplify the colour matrix products:

$$t^a t^c t^a = t^a t^a t^c + t^a [t^c, t^a]$$

Now we use $[t^c, t^a] = i f^{cab} t^b$

$$= t^a t^a t^c + i t^a f^{cab} t^b$$

Now we use $f^{cab} t^a t^b = \frac{1}{2} i C_2(G) t^c$ from Peskin eq(A.36)

$$= C_2(r) t^c - \frac{1}{2} C_2(G) t^c$$

we also have $t^a t^a = C_2(r) \cdot \mathbb{1}$ from Peskin eq (A.34)

$$t^a t^c t^a = \left[C_2(r) - \frac{1}{2} C_2(G)\right] t^c$$

$$\Rightarrow I_1 = i \frac{g^3}{(4\pi)^2} \gamma^\mu \left[C_2(r) - \frac{1}{2} C_2(G)\right] t^c \left[\frac{2}{\varepsilon} - \gamma_E + \ln(4\pi) - \ln\left(\frac{\Lambda}{\mu^2}\right) - \frac{1}{2}\right]$$

So the $\frac{1}{\varepsilon}$ part for vertex-loop  is

$$i \frac{g^3 \gamma^\mu}{(4\pi)^2} \left[C_2(r) - \frac{1}{2} C_2(G) \right] t^c \, \frac{2}{\varepsilon}$$

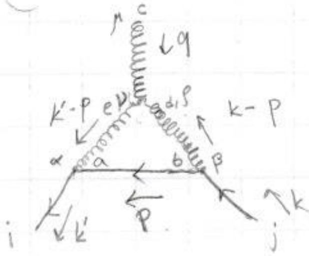
$\Leftarrow$ this is the $\frac{1}{\varepsilon}$ divergent part we were looking for

Here we couldn't also used

$$I_1 = i \frac{g^3}{(4\pi)^2} \left[C_2(r) - \frac{1}{2} C_2(G) \right] t^c \, \Gamma(2 - d/2) \left(1 + \frac{\varepsilon}{2} \ln(4\pi) - \frac{\varepsilon}{2} \ln\left(\frac{\Delta}{\mu^2}\right) - \frac{\varepsilon}{4} + \mathcal{O}(\varepsilon^2) \right)$$

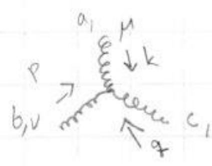
to only extract the divergent $\Gamma(2 - d/2) \cdot 1$ term like in Peskin (16.81)

Now let's calculate the vertex  loop integral



here μ, α, β, μ are Lorentz-indices
a, b, c, d, e are gluon indices
and i, j are colour of external fermions (not needed for loop integral)

The Feynman rule for gluon vertex  is



$$= g f^{abc} \left[g^{\mu\nu} (k-p)^\rho + g^{\nu\rho} (p-q)^\mu + g^{\rho\mu} (q-k)^\nu \right]$$

Using this and the Feynman rules from earlier we get

$$= I_2 = \mu^\varepsilon \int \frac{d^d p}{(2\pi)^d} (i g \gamma^\alpha t^a) \frac{i(\not{p} + m)}{p^2 - m^2} (i g \gamma^\beta t^b) \frac{-i g_{\beta\rho}}{(k-p)^2} \delta_{bd} \, g f^{ced} \left[g^{\mu\nu} (q + k' - p)^\rho \right.$$

$$\left. + g^{\nu\rho} (-k' + p - k + p)^\mu + g^{\rho\mu} (k - p - q)^\nu \right] \frac{-i g_{\alpha\nu}}{(k-p)^2} \delta_{ae}$$

|| Now we set $k' = k = q = m = 0$

$$I_2 = i \mu^\varepsilon g^3 \, t^a f^{cab} t^b \int \frac{d^d p}{(2\pi)^d} \gamma_\rho \not{p} \gamma_\nu \frac{1}{(p^2)^3} \left[-g^{\mu\nu} p^\rho + 2 g^{\nu\rho} p^\mu - g^{\rho\mu} p^\nu \right]$$

$$I_2 = i \mu^\varepsilon g^3 f^{cab} t^a t^b \int \frac{d^d p}{(2\pi)^d} \frac{1}{(p^2)^3} \left[-\gamma_\rho \gamma_\nu \gamma^\mu p^\nu p^\rho + 2 \gamma^\nu \gamma_\rho \gamma_\nu p^\rho p^\mu - \gamma^\mu \gamma_\rho \gamma_\nu p^\rho p^\nu \right]$$

Now we write $p^\mu p^\nu \to \frac{1}{d} g^{\mu\nu} p^2$ just like we did when calculating  -vertex. $\Rightarrow$

$$I_2 = i \mu^\varepsilon g^3 f^{cab} t^a t^b \int \frac{d^d p}{(2\pi)^d} \frac{1}{(p^2)^2} \frac{1}{d} \left[-\gamma_\rho \gamma^\rho \gamma^\mu + 2 \gamma^\nu \gamma^\mu \gamma_\nu - \gamma^\mu \gamma_\rho \gamma^\rho \right]$$

Now we use $\gamma^\mu \gamma_\mu = d$ and $\gamma^\nu \gamma^\mu \gamma_\nu = -(d-2)\gamma^\mu$
from Peskin eq. (A.55)

$$\Rightarrow I_2 = i g^3 \mu^\varepsilon f^{cab} t^a t^b \frac{1}{d} \int \frac{d^d p}{(2\pi)^d} \frac{1}{(p^2)^2} \left(-2d\gamma^\mu - 2(d-2)\gamma^\mu\right)$$

$$= -i g^3 \mu^\varepsilon f^{cab} t^a t^b \frac{1}{d} \int d^d p \frac{1}{(2\pi)^d} \frac{4}{(p^2)^2} (d-1)\gamma^\mu$$

Now let's also use $f^{acb} t^a t^b = \frac{i}{2} C_2(G) t^c$ again.

$$\Rightarrow I_2 = 2 g^3 \mu^\varepsilon C_2(G) t^c \frac{d-1}{d} \gamma^\mu \int \frac{d^d p}{(2\pi)^d} \frac{1}{(p^2)^2}$$

We have already calculated this exact integral using regulator Λ when Calculating the I_1 earlier. Let's use the result from that, along with same approximations.

$$\Rightarrow I_2 = 2 g^3 \mu^\varepsilon C_2(G) t^c \frac{d-1}{d} \gamma^\mu \frac{i}{(4\pi)^{d/2}} \frac{\Gamma(2-d/2)}{\Gamma(2)} \left(\frac{1}{\Lambda}\right)^{2-d/2} \qquad \| \, d = 4-\varepsilon$$

$$\Rightarrow I_2 = i 2 g^3 C_2(G) t^c \gamma^\mu \frac{1}{(4\pi)^2} \Gamma(2-d/2) \left[1 + \frac{\varepsilon}{2}\ln(4\pi) - \frac{\varepsilon}{2}\ln\left(\frac{\Lambda}{\mu^2}\right) + \mathcal{O}(\varepsilon^2)\right] \frac{1}{4}(3-\varepsilon)(1-\varepsilon/4)$$

$$= 2 i g^3 C_2(G) t^c \gamma^\mu \frac{1}{(4\pi)^2} \Gamma(2-\tfrac{d}{2}) \left[1 + \frac{\varepsilon}{2}\ln(4\pi) - \frac{\varepsilon}{2}\ln\left(\frac{\Lambda}{\mu^2}\right) + \mathcal{O}(\varepsilon^2)\right] \left(\frac{3}{4} - \frac{7}{16}\varepsilon + \mathcal{O}(\varepsilon^2)\right)$$

$$I_2 = 2 i g^3 C_2(G) t^c \gamma^\mu \frac{\Gamma(2-d/2)}{(4\pi)^2} \left[\frac{3}{4} + \frac{\varepsilon}{2}\ln(4\pi) - \frac{\varepsilon}{2}\ln\left(\frac{\Lambda}{\mu^2}\right) - \frac{7}{16}\varepsilon + \mathcal{O}(\varepsilon^2)\right]$$

this is a useful form to see $\Gamma(2-d/2)$ - dependence but let's now expand $\Gamma(2-d/2) = \Gamma(\varepsilon/2) = \frac{2}{\varepsilon} - \gamma_E + \mathcal{O}(\varepsilon)$

$$\Rightarrow I_2 = \frac{2i}{(4\pi)^2} g^3 C_2(G) t^c \gamma^\mu \left[\frac{3}{2}\frac{1}{\varepsilon} - \frac{3}{4}\gamma_E + \ln(4\pi) - \ln\left(\frac{\Lambda}{\mu^2}\right) - \frac{7}{8} + \mathcal{O}(\varepsilon)\right]$$

$\Rightarrow$ The $\frac{1}{\varepsilon}$ divergence part of the vertex-loop is :

$$\boxed{3i \frac{g^3}{(4\pi)^2} C_2(G) t^c \gamma^\mu \frac{1}{\varepsilon}}$$

Together the both vertex give the $\frac{1}{\varepsilon}$ contribution of

$$i \frac{g^3}{(4\pi)^2}\left[C_2(r) - \frac{1}{2}C_2(G)\right] t^c \gamma^\mu \frac{2}{\varepsilon} + 3i \frac{g^3}{(4\pi)^2} C_2(G) t^c \gamma^\mu \frac{1}{\varepsilon}$$

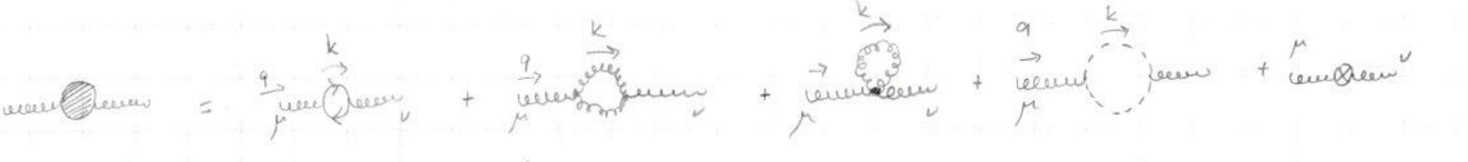
$$\boxed{= i \frac{g^3}{(4\pi)^2}\left[C_2(r) + C_2(G)\right] t^c \gamma^\mu \frac{2}{\varepsilon}}$$

($\frac{1}{\varepsilon}$ - divergence)

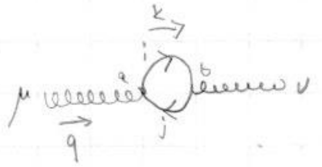
this is the divergent part of the 1-loop contribution to the fermion-gluon vertex.

Exercise 3.9

Compute the divergent part of the gauge boson polarization.



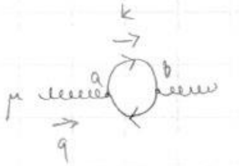
Let's start with Fermion loop:



Because we must take into account all fermion colours we need to take trace of the colour matrix multiplication

The Lorentz structure for gauge-boson self energy is

 $= i\left(q^2 g^{\mu\nu} - q^\mu q^\nu\right) \Pi(q^2)$ due to ward identity

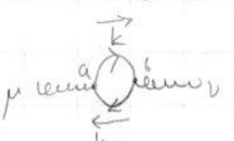
$\Rightarrow$  $= tr\left[t^a t^b\right] i\left(q^2 g^{\mu\nu} - q^\mu q^\nu\right) \Pi_{QED}(q^2)$

where $\Pi_{QED}(q^2)$ is the term that appears in photon self-energy

 $= i\left(q^2 g^{\mu\nu} - q^\mu q^\nu\right)$ and since the only difference between photon polarization and the general gauge boson polarization is the $tr\left[t^a t^b\right]$ from the contribution of group matrix t^a that acts on the fermion gauge group indices, we can use resut from QED. So because In QED the vertex is

 $= ig\gamma^\mu$ with $g = -e$ and in general non-Abelian gauge theory we have  $= ig\gamma^\mu t^a$ with the fermion propagators being same up to the identity matrix in gauge group, we can use result from QED.

So the we use the result from Peskin eq. (7.90) to get $\Pi_{QED}(q^2) = \Pi_2(q^2)$

$\Rightarrow$  $= tr\left[t^a t^b\right] i\left(q^2 g^{\mu\nu} - q^\mu q^\nu\right)\left(\frac{-8g^2}{(4\pi)^{d/2}}\right)\int_0^1 dx \; x(1-x) \frac{\Gamma(2-d/2)}{\Delta^{2-d/2}}$

with $\Delta = m^2 - x(1-x)q^2$ (In peskin the momentums are $k+q$ and k but the result is same since there is only q^2-dependence)

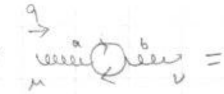
Let's now also use $tr\left[t^a t^b\right] = C(r)\delta^{ab}$ from Peskin eq. (15.78):

$\Rightarrow$  $= C(r) i\left(q^2 g^{\mu\nu} - q^\mu q^\nu\right)\delta^{ab} \frac{-g^2}{(4\pi)^{d/2}}\int_0^1 dx \; 8x(1-x) \frac{\Gamma(2-d/2)}{\left(m^2 - x(1-x)q^2\right)^{2-d/2}}$

Since $d = 4 - \varepsilon$ let's expand the $\dfrac{1}{(m^2 - x(1-x)q^2)^{2-d/2}}$ term to

extract the $\Gamma(2 - d/2)$ divergence:

$$\left[(m^2 - x(1-x)q^2)^{2-(2-\varepsilon/2)}\right]^{-1} = (m^2 - x(1-x)q^2)^{-\varepsilon/2} \approx 1 - \frac{\varepsilon}{2}\ln(m^2 - x(1-x)q^2) + \mathcal{O}(\varepsilon^2)$$

$\Rightarrow$  $= i C(r)(q^2 g^{\mu\nu} - q^\mu q^\nu)\delta^{ab} \dfrac{-g^2}{(4\pi)^{d/2}} \displaystyle\int_0^1 dx\, 8x(1-x)\,\Gamma(2-d/2)\left[1 - \frac{\varepsilon}{2}\ln(m^2 - x(1-x)q^2) + \mathcal{O}(\varepsilon^2)\right]$

Let's also use the approximation $\dfrac{1}{(4\pi)^{d/2}} = \dfrac{1}{(4\pi)^2}\left(1 + \dfrac{\varepsilon}{2}\ln(4\pi) + \mathcal{O}(\varepsilon^2)\right)$

$\Rightarrow$  $= C(r)i(q^2 g^{\mu\nu} - q^\mu q^\nu)\delta^{ab} \dfrac{-g^2}{(4\pi)^2}\displaystyle\int_0^1 dx\, 8x(1-x)\Gamma(2-\frac{d}{2})\left[1 + \frac{\varepsilon}{2}\ln(4\pi) - \frac{\varepsilon}{2}\ln(m^2 - x(1-x)q^2) + \mathcal{O}(\varepsilon^2)\right]$

So the divergent part of this is (since $\Gamma(\varepsilon/2)$ divergent part is of form $\frac{1}{\varepsilon}$)

$$C(r)\, i(q^2 g^{\mu\nu} - q^\mu q^\nu)\delta^{ab}\frac{-g^2}{(4\pi)^2}\int_0^1 dx\, 8x(1-x)\,\Gamma(2 - d/2)$$

$$\int_0^1 dx\; x - x^2 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

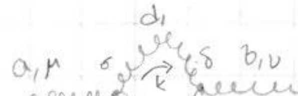
$$\Rightarrow 8 \cdot \frac{1}{6} = \frac{4}{3}$$

$$= i(q^2 g^{\mu\nu} - q^\mu q^\nu)\delta^{ab}\frac{-g^2}{(4\pi)^2}\frac{4}{3}C(r)\,\Gamma(2 - d/2)$$

For N_f fermion species we just sum the contribution by each fermion which amounts to multiplying by N_f; so the divergent part of fermion loop of Gauge polarization is

$$i(q^2 g^{\mu\nu} - q^\mu q^\nu)\delta^{ab}\left(\frac{-g^2}{(4\pi)^2}\frac{4}{3}N_f\, C(r)\,\Gamma(2 - d/2)\right)$$

Now let's calculate the diagram  in Feynman-'t Hooft gauge

 $= \underbrace{\dfrac{1}{2}}_{\text{Symmetry factor}} \displaystyle\int \frac{d^dk}{(2\pi)^d}\frac{-ig_{\sigma\delta}}{k^2}\frac{-ig_{\rho\xi}}{(k-q)^2} g^2 f^{acd} f^{bdc} N^{\mu\nu\sigma\rho\delta\xi}$, where

$$N^{\mu\nu\sigma\rho\delta\xi} = \left[g^{\mu\rho}(q-(k-q))^\sigma + g^{\rho\sigma}((k-q)-(-k))^\mu + g^{\sigma\mu}(-k-q)^\rho\right]\cdot\left[g^{\nu\delta}(-q-k)^\xi + g^{\delta\xi}(k+k-q)^\nu + g^{\nu\xi}(-k+q+q)^\delta\right]$$

where the terms in square brackets come from the 3-gauge boson vertex.

Let's now contract with $g_{\sigma\delta}\, g_{\rho\xi}$ to get $N^{\mu\nu}$

$$g_{\sigma\delta}\, g_{\rho\xi}\, N^{\mu\nu\sigma\rho\delta\xi} = \left[g^{\mu\rho}(2q-k)^\sigma + g^{\rho\sigma}(2k-q)^\mu - g^{\sigma\mu}(k+q)^\rho \right] \left[-\delta^\nu_\sigma (q+k)_\rho \right.$$
$$\left. + g_{\sigma\rho}(2k-q)^\nu + \delta^\nu_\rho (2q-k)_\sigma \right] \equiv N^{\mu\nu}$$

Let's also use Feynman parametrization:

$$\frac{1}{k^2}\frac{1}{(k-q)^2} = \int_0^1 dx \, \frac{1}{\left[(1-x)k^2 + x(k-q)^2\right]^2}$$

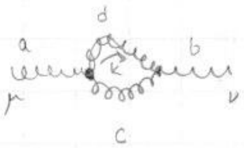
the denominator can be be changed to better form:

$$(1-x)k^2 + x(k-q)^2 = k^2 - xk^2 + xk^2 - 2xkq + xq^2$$

$$= k^2 - 2xkq + x^2q^2 - x^2q^2 + xq^2 \qquad // \text{ shift } l = (k - xq)$$

$$= l^2 + x(1-x)q^2 \qquad // \Delta = -x(1-x)q^2$$

$$= l^2 - \Delta$$



$$= -\frac{g^2}{2} f^{acd} f^{bdc} \int_0^1 dx \int \frac{d^d l}{(2\pi)^d} \frac{1}{(l^2-\Delta)^2} N^{\mu\nu} \qquad // f^{bdc} = -f^{bcd}$$

Peskin eq. (15.93): $f^{acd} f^{bcd} = C_2(G)\,\delta^{ab}$, let's also now write $N^{\mu\nu}$ in terms of l and drop terms linear in l^μ (we drop because the integral is 0 for odd number of l in numerator due to symmetry).

$$N^{\mu\nu} = \left[g^{\mu\rho}(2q-k)^\sigma + g^{\rho\sigma}(2k-q)^\mu - g^{\sigma\mu}(k+q)^\rho \right] \left[-\delta^\nu_\sigma (q+k)_\rho + g_{\sigma\rho}(2k-q)^\nu + \delta^\nu_\rho (2q-k)_\sigma \right]$$

$$N^{\mu\nu} = \left[-(2q-k)^\nu (q+k)^\mu + (2q-k)^\mu (2k-q)^\nu + g^{\mu\nu}(2q-k)^\sigma (2q-k)_\sigma \right. \qquad // g^{\rho\sigma} g_{\sigma\rho} = d$$
$$- (2k-q)^\mu (q+k)^\nu + d(2k-q)^\mu (2k-q)^\nu + (2k-q)^\mu (2q-k)^\nu$$
$$\left. + g^{\mu\nu}(k+q)^\rho (q+k)_\rho - (k+q)^\mu (2k-q)^\nu - (k+q)^\nu (2q-k)^\mu \right]$$

$$N^{\mu\nu} = \left[-(2q-l-xq)^\nu (q+l+xq)^\mu + (2q-l-xq)^\mu (2l+2xq-q)^\nu \right.$$
$$+ g^{\mu\nu}(2q-l-xq)^2 - (2l+2xq-q)^\mu (q+l+xq)^\nu + d(2l+2xq-q)^\mu (2l+2xq-q)^\nu$$
$$+ (2l+2xq-q)^\mu (2q-l-xq)^\nu + g^{\mu\nu}(l+xq+q)^2 - (l+xq+q)^\mu (2l+2xq-q)^\nu$$
$$\left. - (l+xq+q)^\nu (2q-l-xq)^\mu \right]$$

Now we drop odd l - terms

$$\Rightarrow N^{\mu\nu} = \Big[-(2q-xq)^\nu (q+xq)^\mu + \underline{\ell^\nu \ell^\mu} + (2q-xq)^\mu (2xq-q)^\nu \underline{-2\ell^\mu \ell^\nu}$$

$$+ g^{\mu\nu}\left[(2q-xq)^2 + \ell^2\right] - \underline{2\ell^\mu\ell^\nu} - (2xq-q)^\mu (q+xq)^\nu + \underline{4d\ell^\mu\ell^\nu} + d(2xq-q)^\mu(2xq-q)^\nu$$

$$+ (2xq-q)^\mu(2q-xq)^\nu - \underline{2\ell^\mu\ell^\nu} + g^{\mu\nu}\left[\ell^2 + (xq+q)^2\right] - \underline{2\ell^\mu\ell^\nu} - (xq+q)^\mu(2xq-q)^\nu$$

$$+ \underline{\ell^\nu\ell^\mu} - (xq+q)^\nu(2q-xq)^\mu \Big]$$

$$N^{\mu\nu} = \Big[(4d-6)\ell^\mu\ell^\nu + 2g^{\mu\nu}\ell^2 - (2-x)\left[(1+x)+(1+x)\right] q^\mu q^\nu + (2x-1)\Big[d(2x-1)$$

$$+ \underbrace{(2-x) + (2-x) - (1+x) - (1+x)}_{= -2(2x-1) =} \Big] q^\mu q^\nu + g^{\mu\nu}\left[\left[(1+x)^2 + (2-x)^2\right] q^2\right] \Big]$$

$$N^{\mu\nu} = (4d-6)\ell^\mu\ell^\nu + 2g^{\mu\nu}\ell^2 + \left[(2x-1)^2(d-2) - 2(1+x)(2-x)\right] q^\mu q^\nu$$

$$+ g^{\mu\nu} q^2 \left[(2-x)^2 + (1+x)^2\right]$$

Now let's replace $\ell^\mu\ell^\nu \to g^{\mu\nu}\ell^2 \frac{1}{d}$ (by symmetry)

$$\Rightarrow N^{\mu\nu} = g^{\mu\nu} 6\left(1-\tfrac{1}{d}\right)\ell^2 - \underbrace{\left[(2-d)(1-2x)^2 + 2(1+x)(2-x)\right]}_{\equiv A(x)} q^\mu q^\nu$$

$$+ g^{\mu\nu} q^2 \underbrace{\left[(2-x)^2 + (1+x)^2\right]}_{\equiv B(x)}$$

So then we have

$$= \frac{g^2}{2} C_2(G)\,\delta^{ab} \int_0^1 dx \int \frac{d^d\ell}{(2\pi)^d} \frac{g^{\mu\nu}\ell^2 6\left(1-\frac{1}{d}\right) + g^{\mu\nu}q^2 B(x) - A(x) q^\mu q^\nu}{(\ell^2-\Delta)^2}$$

We now use the result for the dimensional integral from Peskin eq. (A.44) and eq. (A.45)

$\Rightarrow$ 

$$= \frac{g^2}{2} C_2(G)\delta^{ab} \int_0^1 dx \Bigg[g^{\mu\nu} 6\left(1-\tfrac{1}{d}\right) \frac{-i}{(4\pi)^{d/2}} \frac{d}{2} \frac{\Gamma(1-d/2)}{\Gamma(2)} \left(\frac{1}{\Delta}\right)^{1-d/2}$$

$$+ \left(g^{\mu\nu} q^2 B(x) - A(x) q^\mu q^\nu\right) \frac{i}{(4\pi)^{d/2}} \frac{\Gamma(2-d/2)}{\Gamma(2)} \left(\frac{1}{\Delta}\right)^{2-d/2} \Bigg]$$

$$\left(\frac{1}{\Delta}\right)^{1-d/2} = \left(\frac{1}{\Delta}\right)^{2-d/2} \cdot \Delta$$

$$\Delta = -x(1-x)q^2$$

$$= \frac{ig^2}{(4\pi)^{d/2}} C_2(G)\delta^{ab} \int_0^1 dx \frac{1}{\Delta^{2-d/2}} \times \Bigg(\Gamma\left(1-\tfrac{d}{2}\right) g^{\mu\nu} q^2 \left[\tfrac{3}{2}(d-1)x(1-x)\right]$$

$$+ \Gamma\left(2-\tfrac{d}{2}\right) g^{\mu\nu} q^2 \left[\tfrac{1}{2}(2-x)^2 + \tfrac{1}{2}(1+x)^2\right] - \Gamma\left(2-\tfrac{d}{2}\right) q^\mu q^\nu \left[\left(1-\tfrac{d}{2}\right)(1-2x)^2 + (1+x)(2-x)\right] \Bigg)$$

$\square$

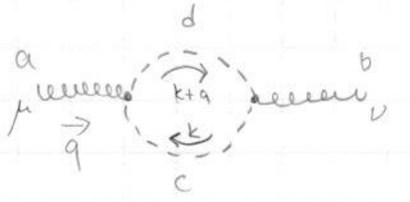
Let's investigate the divergence after we compute the loops

 and . As per the hints we take

 $= 0$ as $\varepsilon \to 0$. Let's now compute the Ghost loop

The Feynman-rules for the ghosts are:

a - - - - - < - - - - b $= \dfrac{i\delta^{ab}}{p^2}$ and  $= -g f^{abc} p^\mu$

 $= (-1) \int \frac{d^d k}{(2\pi)^4} \frac{i}{k^2} \frac{i}{(k+q)^2} g^2 f^{dac} (k+q)^\mu f^{cbd} k^\nu \equiv I_g$

(-1) from anticommutation of Ghost-field

$$\Rightarrow I_g = + f^{dac} f^{cbd} g^2 \int \frac{d^d k}{(2\pi)^4} \frac{1}{k^2} \frac{1}{(k+q)^2} (k+q)^\mu k^\nu$$

Let's now use Feynman parametrization:

$$\frac{1}{k^2}\frac{1}{(k+q)^2} = \int_0^1 dx \frac{1}{[(1-x)k^2 + x(k+q)^2]^2} = \int_0^1 dx \frac{1}{[k^2 + 2xkq + xq^2]^2}$$

Let's shift: $l = k + xq \Rightarrow l^2 = k^2 + 2xkq + x^2q^2 \Rightarrow dk = dl$

$$\Rightarrow \frac{1}{k^2}\frac{1}{(k+q)^2} = \int_0^1 dx \frac{1}{[l^2 - x^2q^2 + xq^2]^2} \qquad \Big\| \; \Delta = x(x-1)q^2$$

$$= \int_0^1 dx \frac{1}{(l^2-\Delta)^2}$$

$$\Rightarrow I_g = f^{dac} f^{cbd} g^2 \int_0^1 dx \int \frac{d^d l}{(2\pi)^4} \frac{1}{(l^2-\Delta)^2} \times \left((k+q)^\mu k^\nu \right)$$

We now make use of the shift $l = k + xq$ in the numerator

$(k+q)^\mu k^\nu = [l - xq + q]^\mu [l - xq]^\nu$ again we drop terms of odd power of l since they contribute 0 to integral due to symmetry

$$\Rightarrow (k+q)^\mu k^\nu = l^\mu l^\nu - x(1-x) q^\mu q^\nu \qquad \Big\| \text{ We also use } l^\mu l^\nu \to \tfrac{1}{d} g^{\mu\nu} l^2 \text{ due to symmetry}$$

$$\Rightarrow I_g = f^{dac} f^{cbd} g^2 \int_0^1 dx \int \frac{d^d l}{(2\pi)^4} \frac{1}{(l^2-\Delta)^2} \left[\frac{1}{d} g^{\mu\nu} l^2 - x(1-x) q^\mu q^\nu \right]$$

Now we use the Minkowski-integrals from Peskin eq. (A.44) and eq. (A.45)

$$\Rightarrow I_g = f^{dac} f^{cbd} g^2 \int_0^1 dx \left[\frac{-i}{(4\pi)^{d/2}} \frac{g^{\mu\nu}}{d} \frac{d}{2} \frac{\Gamma(1-d/2)}{\Gamma(2)} \left(\frac{1}{\Delta}\right)^{1-\frac{d}{2}} - \frac{i}{(4\pi)^{d/2}} \frac{\Gamma(2-d/2)}{\Gamma(2)} \left(\frac{1}{\Delta}\right)^{2-d/2} x(1-x)\, q^\mu q^\nu \right]$$

$$\left(\frac{1}{\Delta}\right)^{1-d/2} = \left(\frac{1}{\Delta}\right)^{2-d/2} \Delta \qquad \| \ \Delta = x(x-1)q^2 = -x(1-x)q^2$$

$$\Rightarrow I_g = \frac{-ig^2}{(4\pi)^{d/2}} f^{dac} f^{cbd} \int_0^1 dx \frac{1}{\Delta^{2-d/2}} \cdot \left(-\Gamma(1-d/2)\, g^{\mu\nu} q^2 \left[\tfrac{1}{2} x(1-x)\right] + \Gamma(2-\tfrac{d}{2})\, q^\mu q^\nu \left[x(1-x)\right] \right)$$

$$\underset{\substack{\uparrow \\ f^{abc} \text{ is totally} \\ \text{anti-symmetric}}}{f^{dac} f^{cbd} = -f^{acd} f^{bcd}} = \underset{\nwarrow \text{ Peskin eq. (15.93)}}{-C_2(G)\,\delta^{ab}}$$

$$\Rightarrow I_g = \frac{ig^2}{(4\pi)^{d/2}} C_2(G)\,\delta^{ab} \int_0^1 dx \frac{1}{\Delta^{2-d/2}} \left(-\Gamma(1-\tfrac{d}{2})\, g^{\mu\nu} q^2 \left(\tfrac{1}{2}x(1-x)\right) + \Gamma(2-\tfrac{d}{2})\, q^\mu q^\nu \left[x(1-x)\right] \right)$$

Now let's add up this with the contribution of 

$\Rightarrow$  $+ I_g \equiv \Sigma$

$$\Rightarrow \Sigma = \frac{ig^2}{(4\pi)^{d/2}} C_2(G)\,\delta^{ab} \int_0^1 dx \frac{1}{\Delta^{2-d/2}} \Bigg[\Gamma(1-\tfrac{d}{2})\, g^{\mu\nu} q^2 \left[x(1-x)\right] \underbrace{\left(\tfrac{3}{2}(d-1) - \tfrac{1}{2}\right)}_{=\frac{3}{2}(d-\frac{4}{3})}$$
$$+ \Gamma(2-\tfrac{d}{2})\, q^\mu q^\nu \left(x(1-x) - (1-\tfrac{d}{2})(1-2x)^2 - (1+x)(2-x) \right)$$
$$+ \Gamma(2-\tfrac{d}{2})\, g^{\mu\nu} q^2 \left(\tfrac{1}{2}(2-x)^2 + \tfrac{1}{2}(1+x)^2 \right) \Bigg]$$

Let's now find the $\frac{1}{\varepsilon}$ divergence, first we note that we can expand $\Gamma(1-d/2)$ around $d=2$ since if we include then this pole is cancelled (Here we don't actually include  since it goes to 0 as $\varepsilon \to 0$, but the pole cancellation is shown in Peskin eq. (16.68)). So we expand $\Gamma(1-d/2)$ around $d=2$ to get $\Gamma(2-d/2)$ term:

$$\Gamma(1-\tfrac{d}{2}) = \frac{1}{1-d/2}\,\Gamma(2-d/2) \qquad \| \text{ Here we use } \Gamma(z+1) = z\,\Gamma(z)$$

$$\Rightarrow \Sigma = \frac{ig^2}{(4\pi)^{d/2}} C_2(G)\,\delta^{ab} \int_0^1 dx \frac{1}{\Delta^{2-d/2}} \Gamma(2-d/2) \Bigg[g^{\mu\nu} q^2 \left[\frac{3}{2} \frac{d-4/3}{1-d/2} x(1-x) + \frac{1}{2}(2-x)^2 + \frac{1}{2}(1+x)^2 \right]$$
$$+ q^\mu q^\nu \left(x(1-x) - (1-\tfrac{d}{2})(1-2x)^2 - (1+x)(2-x) \right) \Bigg]$$

$d = 4-\varepsilon \Rightarrow \frac{d-4/3}{1-d/2} = \frac{4-\varepsilon-4/3}{1-2+\varepsilon/2} = -\left(\frac{8}{3}-\varepsilon\right)\frac{1}{1-\varepsilon/2}$ || Use binomial approximation $(1+\alpha)^x \simeq 1+\alpha x$ for small α

$$\Rightarrow \frac{d-4/3}{1-d/2} = -\left(\frac{8}{3}-\varepsilon\right)\left(1+\varepsilon/2\right) = -\left[\frac{8}{3}+\frac{1}{3}\varepsilon + \mathcal{O}(\varepsilon^2)\right]$$

$$\Rightarrow \frac{3}{2}\frac{d-4/3}{1-d/2} = -\left[4+\frac{\varepsilon}{2}+\mathcal{O}(\varepsilon^2)\right]$$

We also get $\frac{1}{\Delta^{2-d/2}} = \frac{1}{\Delta^{\varepsilon/2}} = \Delta^{-\varepsilon/2} \simeq 1-\frac{\varepsilon}{2}\ln(\Delta)$

$$\frac{1}{(4\pi)^{d/2}} = \frac{1}{(4\pi)^2}(4\pi)^{\varepsilon/2} = \frac{1}{(4\pi)^2}\cdot\left(1+\frac{\varepsilon}{2}\ln(4\pi)\right)$$

We are only interested in $\frac{1}{\varepsilon}$ - term so we may jst write the terms of order ε and higher as $\mathcal{O}(\varepsilon)$

$$\Rightarrow \frac{3}{2}\frac{d-4/3}{1-d/2} = -\left[4+\mathcal{O}(\varepsilon)\right] \quad \text{and} \quad \frac{1}{(4\pi)^{d/2}} = \frac{1}{(4\pi)^2}\left[1+\mathcal{O}(\varepsilon)\right]$$

$$\frac{1}{\Delta^{2-d/2}} = 1+\mathcal{O}(\varepsilon) \quad , \quad 1-\frac{d}{2} = 1-2+\frac{\varepsilon}{2} = -1+\mathcal{O}(\varepsilon)$$

$$\Rightarrow \Sigma = \frac{ig^2}{(4\pi)^2}C_2(G)\delta^{ab}\int_0^1 dx\, \Gamma(2-d/2)\left(1+\mathcal{O}(\varepsilon)\right)\Big[g^{\mu\nu}q^2\Big[-[4+\mathcal{O}(\varepsilon)]\,x(1-x)$$
$$+\frac{1}{2}(2-x)^2+\frac{1}{2}(1+x)^2\Big] + q^\mu q^\nu\left(x(1-x)-(-1+\mathcal{O}(\varepsilon))(1-2x)^2-(1+x)(2-x)\right)\Big]$$

We are only interested in the $\frac{1}{\varepsilon}$ divergence term so we may drop any $\mathcal{O}(\varepsilon)$ terms that appeared in the expression above

$$\Rightarrow \Sigma_{\text{divergence}} = \Sigma_{\text{div}} = \frac{ig^2}{(4\pi)^2}C_2(G)\delta^{ab}\int_0^1 dx\,\Gamma(2-d/2)\Big[g^{\mu\nu}q^2\Big(-4x(1-x)$$
$$+\frac{1}{2}(2-x)^2+\frac{1}{2}(1+x)^2\Big] + q^\mu q^\nu\Big(x(1-x)+(1-2x)^2-(1+x)(2-x)\Big)\Big]$$

Let's first calculate the x-integral before expanding $\Gamma(2-d/2)$ to get the $1/\varepsilon$ - divergence

$$\int_0^1 dx\left[-4x(1-x)+\frac{1}{2}(2-x)^2+\frac{1}{2}(1+x)^2\right] = \frac{5}{3}$$

|| Used calculator to calculate this simple integral to save some time.

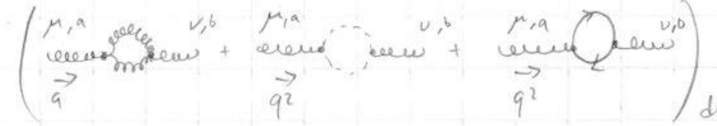
$$\int_0^1 dx\left[x(1-x)+(1-2x)^2-(1+x)(2-x)\right] = -\frac{5}{3}$$

$$\Rightarrow \Sigma_{\text{div}} = \frac{ig^2}{(4\pi)^2}C_2(G)\delta^{ab}\,\Gamma(2-d/2)\left[\frac{5}{3}g^{\mu\nu}q^2-\frac{5}{3}q^\mu q^\nu\right]$$

$$\Sigma_{\text{div}} = i(q^2g^{\mu\nu}-q^\mu q^\nu)\delta^{ab}\left[\frac{-g^2}{(4\pi)^2}\cdot\left(-\frac{5}{3}\right)C_2(G)\,\Gamma(2-d/2)\right]$$

This is the $\Gamma(2-d/2)$ - divergent part of the diagrams

 if we now add the Fermion-loop contribution

of $\Gamma(2-d/2)$-divergence  $\equiv \Pi^{\mu\nu,ab}_{div}(q^2)$

$$\Rightarrow \quad \Pi^{\mu\nu ab}_{div}(q^2) = \Pi^{\mu\nu}_{div}(q^2)\delta^{ab} = i(q^2 g^{\mu\nu} - q^\mu q^\nu)\delta^{ab}\left\{ -\frac{g^2}{(4\pi)^2}\Gamma(2-d/2)\left[\frac{4}{3}C(r)N_f - \frac{5}{3}C_2(G)\right]\right\}$$

So the 1-loop divergence of gauge boson polarization is

$$i(q^2 g^{\mu\nu} - q^\mu q^\nu)\delta^{ab}\left\{ +\frac{g^2}{(4\pi)^2}\left[\frac{5}{3}C_2(G) - \frac{4}{3}C(r)N_f\right]\Gamma(2-d/2)\right\}$$

To extract the $1/\varepsilon$ divergence we expand $\Gamma(2-d/2) = \Gamma(\varepsilon/2) = \frac{2}{\varepsilon} - \gamma_E + \mathcal{O}(\varepsilon)$ and drop the non $1/\varepsilon$ - terms.

$\Rightarrow$ The $1/\varepsilon$ divergence of the gauge boson polarization is

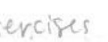
$$\boxed{i(q^2 g^{\mu\nu} - q^\mu q^\nu)\delta^{ab}\left[\frac{g^2}{(4\pi)^2}\left[\frac{5}{3}C_2(G) - \frac{4}{3}C(r)N_f\right]\right]\frac{2}{\varepsilon}}$$

Exercise 3.10

We get the counter terms from previous exercises result

δ_3 is the counter term for gauge boson polarization, we get this from ex. 3.9 result

δ_1 is the counter term for vertex correction, we get this from ex. 3.8 result

δ_2 is the counter term for field-strenght , we get this from ex. 3.7 result from lecture notes.

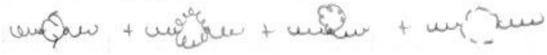
We will be working in MS-scheme

The counter term δ_2 is $\delta_2^{MS} = -\frac{g^2}{8\pi^2}C_2(r)\frac{1}{\varepsilon}$

δ_3 an δ_2 we need to calculate from results of ex. 3.9 and 3.8

Let's start with δ_1

we get δ_3 from $+i\Pi =$  $- i(g^{\mu\nu}q^2 - q^\mu q^\nu)\delta_3\delta^{ab}$

In MS-scheme we cancel only $\frac{1}{\varepsilon}$ term with counter term and we got the $1/\varepsilon$ term for  to be

$$i(q^2 g^{\mu\nu} - q^\mu q^\nu)\,\delta^{ab}\left(\frac{g^2}{(4\pi)^2}\left(\frac{5}{3}C_2(G) - \frac{4}{3}C(r)N_f\right)\right)\frac{2}{\varepsilon}$$

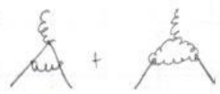
this gives the counter term

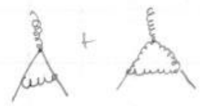
$$\delta_3^{MS} = \frac{g^2}{(4\pi)^2}\left[\frac{5}{3}C_2(G) - \frac{4}{3}C(r)N_f\right]\frac{2}{\varepsilon}$$

$$\delta_3^{MS} = \frac{g^2}{16\pi^2}\left[\frac{10}{3}C_2(G) - \frac{8}{3}C(r)N_f\right]\frac{1}{\varepsilon}$$

Now we compute δ_1^{MS} from result of exercise 3.8

The counter term is given by

$-i\Gamma =$  $+\, i g t^a \gamma^\mu \delta_1$

We have $\frac{1}{\varepsilon}$-divergence of  to be (from exercise 3.8)

$$i\frac{g^3}{(4\pi)^2}\left[C_2(r) + C_2(G)\right]t^c\gamma^\mu\frac{2}{\varepsilon}$$, this gives the counter term

$$\delta_1^{MS} = -\frac{g^2}{16\pi^2}\left[C_2(r) + C_2(G)\right]\frac{2}{\varepsilon}$$

So now we have $\delta_1^{MS} = -\frac{g^2}{8\pi^2}\left[C_2(r) + C_2(G)\right]\frac{1}{\varepsilon}$

$\delta_2^{MS} = -\frac{g^2}{8\pi^2}C_2(r)\frac{1}{\varepsilon}$, $\delta_3^{MS} = \frac{g^2}{8\pi^2}\left[\frac{5}{3}C_2(G) - \frac{4}{3}C(r)N_f\right]\frac{1}{\varepsilon}$

To the 1-loop order β-function is scheme independent:

$g_0 = Z_1 Z_2^{-1} Z_3^{-1/2}\mu^{\varepsilon/2} g = Z_g \mu^{\varepsilon/2} g$, from scale independence of g_0 we get

$$0 = \frac{\partial g_0}{\partial\mu} = \frac{\varepsilon}{2}Z_g g\,\mu^{\varepsilon/2 - 1} + \mu^{\varepsilon/2}\frac{\partial}{\partial\mu}(Z_g g)$$

, From this we get

$$\beta_0(g) = \frac{g^2}{2}\frac{dZ_{g,1}}{dg}$$

, here $Z_{g,1}$ is the part of Z_g but without the $\frac{1}{\varepsilon}$ dependence since it comes from expansion $Z_g = 1 + \sum_{n=1}^{\infty}\frac{Z_{g,n}(g)}{\varepsilon^n}$

The derivation from $\frac{\partial g_0}{\partial\mu} = 0$ to $\beta_0(g) = \frac{g^2}{2}\frac{dZ_{g,1}}{dg}$ is identical to the one in QFT 3 week 6 derivation of $\beta_0(g)$ so we omit the derivation here. Let's now continue from the $\beta_0(g) = \frac{g^2}{2}\frac{dZ_{g,1}}{dg}$

In 1-loop order:

$$\beta(g) = \beta_0(g) = \frac{g^2}{2}\frac{dZ_{g,1}}{dg}$$, now $Z_{g,1} = Z_{1,1} Z_{2,1}^{-1} Z_{3,1}^{-1/2}$

such that $Z_{i,1} = 1 + \varepsilon\delta_i$.

Let's write $\varepsilon\delta_i \to \delta_i$ so that δ_i is just counter-term without $\frac{1}{\varepsilon}$ part

now we get:

$$\Rightarrow \beta(g) = \frac{g^2}{2}\frac{d}{dg}\left[\frac{1+\delta_1}{(1+\delta_2)(1+\delta_3)^{1/2}}\right]$$

expand in terms of small g^2 since $\delta_i \sim g^2$

$$\beta(g) = \frac{g^2}{2}\frac{d}{dg}\left[(1+\delta_1)\left[(1+\delta_2)(1+\tfrac{1}{2}\delta_3)\right]^{-1}\right]$$

binomial approximation:

$(1+\delta_3)^{1/2} \approx 1 + \frac{1}{2}\delta_3$

we set $\mathcal{O}(g^4) = 0$

since g^2 = small

$$= \frac{g^2}{2}\frac{d}{dg}\left[(1+\delta_1)\left[1+\delta_2+\tfrac{1}{2}\delta_3\right]^{-1}\right]$$

Again binomial approximation

$$= -\frac{g^2}{2}\frac{d}{dg}\left[-(1+\delta_1)\left(1-\delta_2-\tfrac{1}{2}\delta_3\right)\right]$$

Now we substitute in δ_1, δ_2, δ_3

(no $1/\varepsilon$ terms)

$$= -\frac{g^2}{2}\frac{d}{dg}\left[-\delta_1+\delta_2+\tfrac{1}{2}\delta_3 - 1\right]$$

$\frac{d}{dg}(-1) = 0$

$$= -\frac{g^2}{2}\frac{d}{dg}\left[\frac{g^2}{8\pi^2}\left(\cancel{C_2(r)} + C_2(G) - \cancel{C_2(r)} + \frac{1}{2}\left[\frac{5}{3}C_2(G) - \frac{4}{3}C(r)N_f\right]\right)\right]$$

$$= -\frac{g^3}{8\pi^2}\left[\frac{11}{6}C_2(G) - \frac{4}{6}C(r)N_f\right]$$

For $SU(N)$, $G = N$

$C_2(N) = N$ and

$C(r) = \frac{1}{2}$

$$\beta(g) = -\frac{g^3}{48\pi^2}\left[11N - 2N_f\right] \quad \square$$

We get the correct $\beta(g)$. In this exercis only thing I'm unsure about is the way I handeled $Z_{g,1}$, but it should be fine since lecture notes also $Z_{g,1}$ is coeff of $1/\varepsilon$ in Z_g but then I'm wondering if it's supposed to be $1+\varepsilon\delta_i$ or just $\varepsilon\delta_i$, with δ_i being the counter term with $1/\varepsilon$ included. $1+\varepsilon\delta_i$ work and it is what I've used, tho it might be wrongly done since I expanded around weak coupling g.